André Pascoal Bento

Platform Engineer @ Mollie | Invited Assistant Professor @ University of Coimbra | Ph.D. in Informatics Engineering

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andre-bento CISUC Profile Scholar andrepbento
Communication skills: Portuguese (native), English (C1 - advanced), French (B1 - intermediate), Spanish (B1 - intermediate).

About Platform Engineer at Mollie with a Ph.D. in Informatics Engineering from the University of Coimbra, where I also teach as an Invited Assistant Professor. My academic and industry work focuses on optimizing the availability and resource efficiency of cloud services, blending practical engineering with research-driven insights. I hold a M.Sc. from the University of Coimbra, and a B.Sc. from the Coimbra Institute of Engineering. Passionate about distributed systems and cloud infrastructure, I continuously seek opportunities to build reliable, scalable systems and deepen my expertise. In my free time, outside of work, I enjoy swimming, cycling, exploring nature, and playing the guitar.

Experience

Aug 2025 - Present, *Platform Engineer*, Mollie
Member of the Reliability team, involved in company-wide incident reviews and on-call tooling (Datadog, PagerDuty, Sentry). Own the observability data pipelines responsible for company-wide metrics ingestion, retention, and filtering using Vector (Datadog), running on Kubernetes on Google Cloud. Led the completion of a metrics filtering pipeline that improved governance over metrics and delivered substantial cost savings.

Sep 2021 - Present, *Invited Assistant Professor*, University of Coimbra
Teaching practical laboratory classes on Distributed Systems (B.Sc.) and Systems Integration (M.Sc.) in Informatics Engineering. Topics in Distributed Systems include multi-threading, parallel programming, RMI/RPC, REST services (Spring Boot and FastAPI), and WebSockets. Systems Integration topics cover microservices, service-oriented architectures, integration patterns, data serialization formats (XML, JSON, and Protocol Buffers), Reactive REST APIs and event-driven systems (Apache Kafka). Also taught Introduction to Programming (B.Sc.) in the 2025/2026 school year.

Sep 2019 - Present, *Researcher*, CISUC - Centre for Informatics and Systems of the University of Coimbra
Researching optimization techniques and root-cause analysis to improve availability and resource utilization of cloud services. Developing solutions using Docker, Kubernetes, Helm, Terraform, Ansible, CI/CD and AWS, alongside service mesh (Istio), observability tools (Grafana, Prometheus, and Jaeger) and data analysis using Pandas, NumPy, and SciPy. Utilizing Python, Java, and Go as programming languages and Git as a version control system.

Sep 2018 - Jul 2019, *Research Intern*, CISUC - Centre for Informatics and Systems of the University of Coimbra
Researched microservices, observability and performance monitoring using metrics, logs, and distributed tracing.

Feb 2017 - Jul 2017, *Software Engineer Intern*, WIT Software, S.A.
Developed a Mobile Augmented Reality prototype for Android and iOS, implementing digital filters, image manipulation, and user content creation features (*e.g.*, selfies, stickers, photo effects, sound effects, emojis, and drawings).

Education

2019 - 2025, Ph.D. in Informatics Engineering, University of Coimbra
Thesis: Optimizing Availability and Resource Utilization of Cloud Services.

2017 - 2019, M.Sc. in Informatics Engineering, University of Coimbra
Thesis: Observing and Controlling Performance in Microservices.

2014 - 2017, B.Sc. in Informatics Engineering, Coimbra Institute of Engineering (ISEC)

Grants and Projects

- **Dec 2021 - Jul 2025, Ph.D. grant** - Foundation for Science and Technology (FCT) grant number BD.06012.2021.
- **2023 - 2025, ROAR-NET** - Randomised Optimisation Algorithms Research Network (CA22137). Funded by COST.
- **2019 - 2022, AESOP** - Autonomic Service Operation. Funded by P2020-31/SI/2017, No. 040004.

Publications

1. **Andre Bento**, Filipe Araujo, Luís Paquete, and Raul Barbosa. *Optimal Scaling of Cloud Services*. In IEEE Transactions on Cloud Computing.
2. **Andre Bento**, Filipe Araujo, and Raul Barbosa. *Cost-availability aware scaling: Towards optimal scaling of cloud services*. Journal of Grid Computing, 21(4):80, 2023.
3. Gonçalo Baptista, Jaime Correia, **Andre Bento**, Joao Soares, Antonio Ferreira, Joao Duraes, Raul Barbosa, and Filipe Araujo. *Defektor: An extensible tool for fault injection campaign management in microservice systems*. In Proceedings of the 38th ACM/SIGAPP Symposium on Applied Computing, SAC'23, page 184-187, New York, NY, USA, 2023.
4. Stanley Lima, Filipe Araujo, Miguel de Oliveira Guerreiro, Jaime Correia, **Andre Bento**, and Raul Barbosa. *Efficient causal access in geo-replicated storage systems*. Journal of Grid Computing, 21(1):8, 2023.
5. **Andre Bento**, Joao Soares, António Ferreira, Joao Duraes, José Ferreira, Rita Carreira, Filipe Araujo, and Raul Barbosa. *Bi-objective optimization of availability and cost for cloud services*. In 2022 IEEE 21st International Symposium on Network Computing and Applications (NCA), volume 21, pages 45-53. IEEE, 2022.
6. **Andre Bento**, Jaime Correia, Joao Duraes, João Soares, Luís Ribeiro, António Ferreira, Rita Carreira, Filipe Araujo, and Raul Barbosa. *A layered framework for root cause diagnosis of microservices*. In 2021 IEEE 20th International Symposium on Network Computing and Applications (NCA), pages 1-8. IEEE, 2021.
7. João Tomás, **Andre Bento**, João Soares, Luís Ribeiro, António Ferreira, Rita Carreira, Filipe Araújo, and Raul Barbosa. *Autonomic service operation for cloud applications: Safe actuation and risk management*. In Dependable Computing-EDCC 2021 Workshops: DREAMS, DSOGRI, SERENE 2021, Munich, Germany, September 13, 2021, Proceedings 17, pages 39-46. Springer, 2021.
8. Sara Silva, Jaime Correia, **Andre Bento**, Filipe Araujo, and Raul Barbosa. *μViz: Visualization of microservices*. In 2021 25th International Conference Information Visualisation (IV), pages 120-128. IEEE, 2021.
9. **Andre Bento**, Jaime Correia, Ricardo Filipe, Filipe Araujo, and Jorge Cardoso. *Automated analysis of distributed tracing: Challenges and research directions*. Journal of Grid Computing, 19:1-15, 2021.